

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Operation Systems

COURSE CODE: CSA0409

COURSE OUTCOME COVERED

CO4: Optimize various file allocation strategies and disk scheduling algorithms to identify optimal methods for enhancing system performance.

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

CODING CHALLENGE / HACKATHON

Code of Conduct:

I k.jaya venkata sai / 192525182 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

k.jaya venkata sai

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main.c

```
1 int main()
2 {
3     int n, start, size, i, index;
4
5     printf("Enter total blocks: ");
6     scanf("%d", &n);
7
8     printf("Enter starting block: ");
9     scanf("%d", &start);
10
11    printf("Enter file size: ");
12    scanf("%d", &size);
13
14    printf("\nContiguous Allocation:\n");
15    for(i = start; i < start + size; i++)
16        printf("%d ", i);
17
18    printf("\n\nLinked Allocation:\n");
19    for(i = start; i < start + size; i++)
20        printf("%d -> ", i);
21    printf("NULL");
22
23
24    printf("\n\nEnter index block: ");
25    scanf("%d", &index);
26
27    printf("Indexed Allocation:\n");
28    printf("Index Block: %d\n", index);
29
30    for(i = 0; i < size; i++)
31        printf("%d ", i);
32
33    return 0;
34 }
```



input

5 -> 6 -> 7 -> 8 -> NULL

Enter index block: 2

Indexed Allocation:

Index Block: 2

0 1 2 3

...Program finished with exit code 0

Press ENTER to exit console.

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main.c

```
5
6     printf("Enter number of blocks: ");
7     scanf("%d", &n);
8
9     int bitmap[n];
10
11     printf("Enter bitmap (0=free, 1=allocated):\n");
12     for(i = 0; i < n; i++)
13         scanf("%d", &bitmap[i]);
14
15     printf("Enter blocks required: ");
16     scanf("%d", &need);
17
18     for(i = 0; i < n && count < need; i++) {
19         if(bitmap[i] == 0) {
20             bitmap[i] = 1;
21             printf("Block %d allocated\n", i);
22             count++;
23         }
24     }
25
26     if(count < need)
27         printf("Allocation failed\n");
28     else
29         printf("Allocation successful\n");
30
31     printf("Updated Bitmap: ");
32     for(i = 0; i < n; i++)
33         printf("%d ", bitmap[i]);
34
35     return 0;
36 }
```

<

input

```
Enter blocks required: 3
Block 1 allocated
Block 3 allocated
Block 4 allocated
Allocation successful
Updated Bitmap: 1 1 1 1 1 1 0 0
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

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main.c

```
22     fcfs += abs(current - a[i]),
23     current = a[i];
24 }
25
26 /* SSTF */
27 current = head;
28
29 for(i = 0; i < n; i++) {
30     min = 9999;
31
32     for(int j = 0; j < n; j++) {
33         if(!visited[j] && abs(current - a[j]) < min) {
34             min = abs(current - a[j]);
35             pos = j;
36         }
37     }
38
39     visited[pos] = 1;
40     sstf += min;
41     current = a[pos];
42 }
43
44 printf("\nFCFS Head Movement = %d", fcfs);
45 printf("\nSSTF Head Movement = %d", sstf);
46
47 if(fcfs < sstf)
48     printf("\nBest Algorithm = FCFS");
49 else
50     printf("\nBest Algorithm = SSTF");
51
52 return 0;
53 }
```



input

```
Enter requests: 82 170 43 140 24
Enter initial head: 50
```

```
FCFS Head Movement = 460
SSTF Head Movement = 172
Best Algorithm = SSTF
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```



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main.c

```
40
41  /* Move to end and beginning */
42  cscan += size - 1 - head;
43  cscan += size - 1;
44
45  /* LOOK */
46  head = 50;
47
48  for(i = 0; i < n; i++) {
49      if(a[i] >= head) {
50          look += abs(head - a[i]);
51          head = a[i];
52      }
53  }
54
55  for(i = n - 1; i >= 0; i--) {
56      if(a[i] < 50) {
57          look += abs(head - a[i]);
58          head = a[i];
59      }
60  }
61
62  printf("\nC-SCAN Head Movement = %d", cscan);
63  printf("\nLOOK Head Movement = %d", look);
64
65  if(cscan < look)
66      printf("\nBest Algorithm = C-SCAN");
67  else
68      printf("\nBest Algorithm = LOOK");
69
70  return 0;
71 }
```

input

Enter initial head: 50

Enter disk size: 200

C-SCAN Head Movement = 348

LOOK Head Movement = 266

Best Algorithm = LOOK

...Program finished with exit code 0

Press ENTER to exit console.



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main.c

```
10
11     printf("Enter starting block: ");
12     scanf("%d", &start);
13
14     printf("Enter file size: ");
15     scanf("%d", &size);
16
17     printf("Enter initial head: ");
18     scanf("%d", &head);
19
20     printf("\nAllocated Blocks: ");
21
22     for(i = 0; i < size; i++) {
23         block[i] = start + i;
24         printf("%d ", block[i]);
25     }
26
27     /* FCFS disk scheduling */
28     for(i = 0; i < size; i++) {
29         movement += abs(head - block[i]);
30         head = block[i];
31     }
32
33     printf("\n\nFCFS Sequence: ");
34
35     for(i = 0; i < size; i++)
36         printf("%d ", block[i]);
37
38     printf("\nTotal Head Movement = %d", movement);
39
40     return 0;
41 }
```

input

Enter initial head: 3

Allocated Blocks: 5 6 7 8

FCFS Sequence: 5 6 7 8

Total Head Movement = 5

...Program finished with exit code 0

Press ENTER to exit console.